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Semester: III	**Year : AY 2026-27**
Subject: Programming with Java	
ASSIGNMENT No : 15	
TITLE : Exception Handling using Finally, Throw, and Throws	

Problem Statement

Write a Java program to demonstrate exception handling using finally, throw, and throws.

Learning Objectives

To implement finally, throw, and throws for effective exception management.

Programme

1. **Create a Login program that throws an exception for invalid password and uses finally block.**

Code:

```
import java.util.Scanner;

public class Main {
   static void login(String password) throws Exception {
      if (!password.equals("12345")) {
         throw new Exception("Invalid password!");
      }
      System.out.println("Login successful.");
   }

   public static void main(String[] args) {
      Scanner sc = new Scanner(System.in);

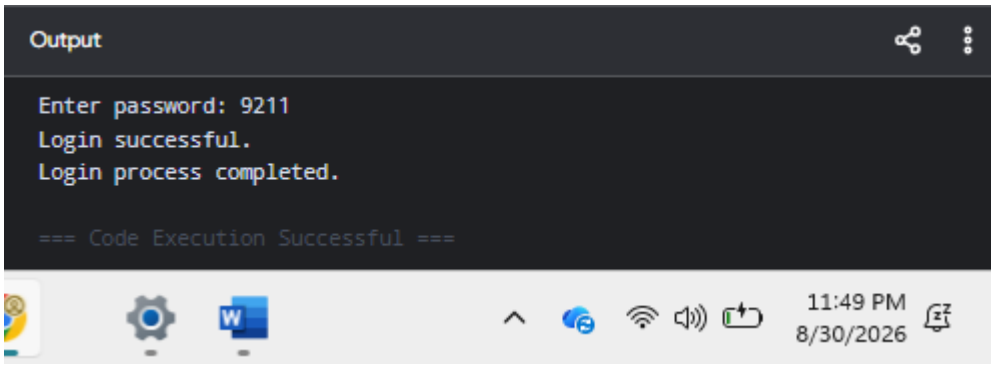
      System.out.print("Enter password: ");
      String password = sc.nextLine();

      try {
         login(password);
      } catch (Exception e) {
         System.out.println("Error: " + e.getMessage());
```

```
        } finally {
            System.out.println("Login process completed.");
        }

        sc.close();
    }
}
```

Output Screenshot



2. **Create an ATM PIN Verification program that throws an exception for an invalid PIN entered by the user and uses a finally block to display a message indicating that the verification process has completed.**

Code:

```
import java.util.Scanner;

public class Main {
    static void verifyPIN(int pin) throws Exception {
        if (pin != 1234) {
            throw new Exception("Invalid PIN!");
        }
        System.out.println("PIN verified successfully.");
    }

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

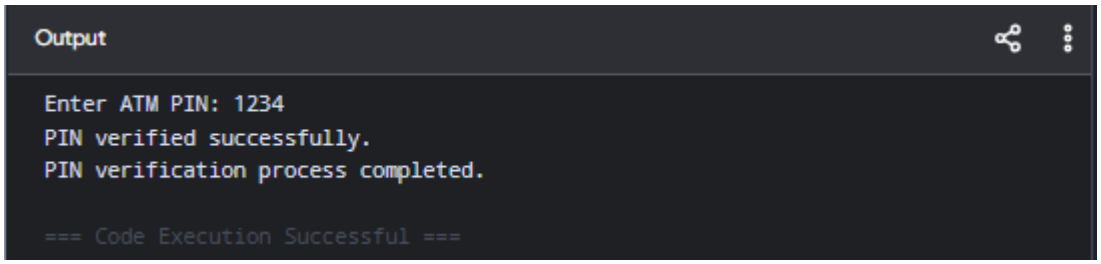
        System.out.print("Enter ATM PIN: ");
        int pin = sc.nextInt();

        try {
            verifyPIN(pin);
        } catch (Exception e) {
            System.out.println("Error: " + e.getMessage());
```

```
        } finally {
            System.out.println("PIN verification process completed.");
        }

        sc.close();
    }
}
```

Output Screenshot:



Conclusion

The programs demonstrate the use of finally, throw, and throws for effective exception management in Java. The Login program throws an exception when an invalid password is entered, while the ATM PIN Verification program handles an invalid PIN using the same concept. The throw keyword is used to generate an exception explicitly, and throws declares that a method may pass an exception to its caller. The finally block executes regardless of whether an exception occurs and displays the completion message. These mechanisms make exception handling more structured, reliable, and user-friendly.